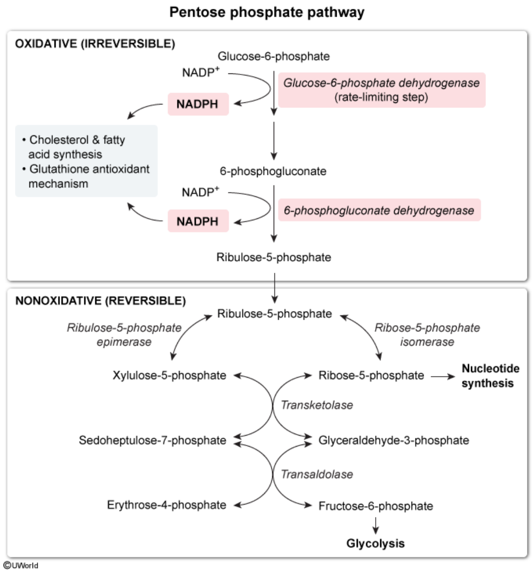
Which two enzymes of the pentose phosphate pathway catalyze production of NADPH?

- A. Ribulose 5-phosphate epimerase and ribose 5-phosphate isomerase [23%]
- B. Transketolase and transaldolase [3%]
- C. Glucose 6-phosphate dehydrogenase and 6-phosphogluconate dehydrogenase [71%]
- D. Lactonase and aldolase [1%]

Correct
71% answered correctly

Explanation:



The **pentose phosphate pathway** consumes glucose 6-phosphate to produce **ribose 5-phosphate** for nucleotide synthesis and **NADPH** for use in lipid synthesis and antioxidative processes. Excess ribose 5-phosphate is converted into the glycolysis intermediates fructose 6-phosphate and glyceraldehyde 3-phosphate. These intermediates can then be recycled back into the pentose phosphate pathway to produce more NADPH, or into glycolysis to generate ATP.

Oxidoreductases interconvert NAD^+ and NADH, $NADP^+$ and NADPH, or FAD and $FADH_2$ by transferring hydride ions or hydrogen atoms. They frequently have the word **"dehydrogenase"** in their names. **Glucose 6-phosphate dehydrogenase** and **6-phosphogluconate dehydrogenase** are the only oxidoreductases in the pentose phosphate pathway; they transfer hydride ions from glucose 6-phosphate and 6-phosphogluconate to $NADP^+$. These enzymes generate the majority of the NADPH in a cell.

(Choice A) Ribose 5-phosphate isomerase and ribulose 5-phosphate epimerase isomerize ribulose 5-phosphate to ribose 5-phosphate and xylulose 5-phosphate, respectively. They occur in the nonoxidative phase of the pentose phosphate pathway and do not generate NADPH.

(Choice B) Transketolase and transaldolase transfer carbons from one sugar phosphate to another, ultimately regenerating the glycolysis intermediates fructose 6-phosphate and glyceraldehyde 3-phosphate. They do not participate in NADPH synthesis.

(Choice D) Lactonase hydrolyzes 6-phosphogluconolactone to form 6-phosphogluconate without generating NADPH. Aldolase is part of glycolysis and gluconeogenesis, but not the pentose phosphate pathway.

Educational objective:
The pentose phosphate pathway generates ribose 5-phosphate and NADPH. The NADPH is generated by reduction of $NADP^+$, which is catalyzed by the oxidoreductases glucose 6-phosphate dehydrogenase and 6-phosphogluconate dehydrogenase.

Time spent: 0 sec	**QID:** 401776
Subject: Biochemistry	**System:** Metabolic Reactions